



Sustainability-Based VOE Governance Model in Pursuit of Resilient Village Development from the Economic and Environmental Perspectives

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Abstract: This study developed a sustainability-based governance model for Village-Owned Enterprises (VOEs) to support the acceleration of sustainable development at the village level. Weak governance structures, limited human resource capacity, and insufficient integration of sustainability values continue to constrain VOE effectiveness as drivers of local economic development. Having used the Analytic Hierarchy Process (AHP), this study evaluated five governance criteria: Human resource capacity and quality, transparency and accountability, collaboration and partnership, environmental commitment, as well as community participation and empowerment, across six sustainability-oriented VOEs. The results indicated that human resource capacity and quality constituted the highest priority (weight 0.3333), followed by transparency and accountability (0.2667) and cross-sector collaboration (0.2000). Although environmental commitment and community participation received lower priority weights, evidence from a qualitative study demonstrated that these dimensions played a critical role in strengthening socio-ecological resilience. Empirical cases from Ponggok and Kenteng VOEs showed that water conservation initiatives and waste management innovations were essential to sustaining long-term economic performance. Overall, the findings suggested that effective VOE governance extended beyond administrative functions toward a transformative model that integrated institutional capacity, social legitimacy, and environmental stewardship. This governance framework positions VOEs as socio-ecological actors contributing to the achievement of Sustainable Development Goals (SDGs), thus highlighting sustainability-oriented governance as a strategic prerequisite for resilient village development in the context of green transition.

Keywords: Governance; Sustainability; Village-Owned Enterprises; Analytic Hierarchy Process

1. Introduction

The crucial role of villages in accelerating development in Indonesia is significant, given that Indonesia has 74,960 villages, and approximately 71% of its population lives in rural areas (Mahrofi, 2022). If village development is implemented effectively, villages can become a significant driving force in advancing national development, for example, by empowering the rural economy (Naudé, 2010). A rural economy is defined as business activities operating in rural areas that are distinct from those in urban or metropolitan areas (Covin & Slevin, 1991). This concept encompasses a wide range of economic activities in rural areas, including agriculture, agribusiness, tourism, crafts, and other sectors.

Economic empowerment at the village level in Indonesia has been realized and encouraged through business optimization implemented through Village-Owned Enterprises (VOEs) (Kania et al., 2021). Thus, VOEs can be a tool for village development while alleviating poverty (Chatterjee et al., 2017; Srijekti, 2018; Steiner & Atterton, 2015). VOEs, as the axis of the village economy, require good governance so that they can provide a significant multiplier effect for the village community in general, and the village government in particular. By holding a legal entity status (Commanditaire Vennotschap or Perseroan Terbatas), business governance and management can run smoothly. However, existing challenges, such as massive digitalization, human resource capacity, climate change,

and environmental impacts on ecosystems, require special attention.

Top-down management stifles business creativity and innovation, leading to subpar performance. Profit sharing between VOs and the village government, which is included in the Village Original Income, should be in place. The Ministry of Villages, Development of Disadvantaged Regions, and Transmigration (Kemendesa) periodically monitors and evaluates the performance of VOs. According to the data from the Ministry of Villages, 21,901 VOs in Indonesia are legally registered and actively conducting business activities. The evaluation identified four criteria for VO evolution: Foundation, Growth, Development, and Progress. Effective management governance plays a crucial role in the support of business sustainability.

In the long run, business sustainability can create prosperous and economically independent communities. This encourages and fosters sustainable village development. The Financial and Development Supervisory Agency (FDSA) explained that the causes of poor governance included suboptimal business planning and the unfavourable development of village potential or unique resources. Difficulties of marketing arise from an inability to synergize with third parties, and financial reporting that does not comply with regulations (Steiner & Atterton, 2015). If these VO issues can be addressed, the growth and development of VOs can lead to sustainable business outcomes. Furthermore, VO development could enhance the village economy by promoting the growth of village industries. Business institutions also provide certainty in carrying out business activities, such as applying for capital (Susilowati et al., 2021), thereby creating a broad market expansion.

The crisis of climate change impacts business activities and requires managers to consider environmental aspects. Businesses should not neglect, disrupt, or damage the environment, given that most VOs operate in the fields of water, waste, and tourism management. Numerous studies examined the concept of sustainability, such as green economy in the agricultural sector (Pangarso et al., 2022), tourism sector (Musvoto et al., 2018; Nugroho et al., 2023), industrial sector, and even at the macro-economic level (Handayani et al., 2022). Therefore, a sustainability-based VOs governance model is required to support sustainable village development.

The urgency of this research lay in developing a business governance model that provided balanced environmental and business sustainability benefits, thereby supporting sustainable village development. It supported the Village Sustainable Development Goals (SDGs), specifically number 8, equitable village economic growth, and number 12, environmentally conscious village consumption and production. Furthermore, Voumik et al. (2023) stated that green economy-based businesses were the main driving force for achieving economic, social, and environmental sustainability. To ensure future growth and sustainable development, sound environmental quality is required (Chaaben et al., 2024).

Despite the expanding literature on community-based enterprises and local economic institutions, prior studies on sustainability governance predominantly focused on corporate organizations, institutions in the public sector, and national policy frameworks, leaving subnational and community-owned enterprises underexplored. Research on sustainability governance has emphasized the importance of human capital, accountability, stakeholder engagement, and institutional coordination in advancing sustainable development (Bansal & Song, 2017; Benn et al., 2014; Schaltegger et al., 2019). Similarly, studies on the implementation of SDGs have largely examined governance mechanisms at the macro- or meso-level, such as national governments, multinational corporations, and global partnerships (Biermann et al., 2017; United Nations Development Programme, 2009). However, there is limited empirical evidence explaining how sustainability principles were operationalized through internal governance mechanisms at the village or community enterprise level.

Within the specific context of village-owned or community-owned enterprises, existing studies have primarily addressed issues of institutional legitimacy, financial performance, and administrative compliance, often treating governance dimensions and environmental concerns as separate analytical domains (Battilana et al., 2015; Doherty et al., 2014; Peredo & Chrisman, 2006; Schmiedeknecht, 2019). While recent studies have acknowledged the potential of social and community enterprises to contribute to inclusive and sustainable development, few studies have systematically integrated governance criteria, such as human resource capacity, transparency and accountability, collaboration networks, environmental commitment, and community participation, into a coherent sustainability-oriented governance model (Benn et al., 2014; Biermann et al., 2017; Quaye et al., 2026). Addressing this gap, the present study advanced the governance and sustainability literature by proposing a sustainability-based governance model for VOs using the Analytic Hierarchy Process (AHP) to prioritize governance dimensions linked to SDGs implementation. By doing so, this study contributed beyond the VO context to offer transferable insights into how locally embedded and community-owned enterprises could serve as strategic governance actors to advance sustainable development at the grassroots level, particularly in developing and emerging economies.

While this study proposed an integrated analytical framework combining AHP, participation theory, and collaborative governance theory, their logical relationship required further clarification. These three components operate at different yet interconnected governance levels. First, AHP functions as a decision-support instrument that structures and prioritises governance criteria based on experts' judgment. As a technocratic tool, AHP does not replace democratic processes but provides a systematic mechanism to reduce ambiguity in complex decision environments. In sustainability governance literature, structured prioritisation tools are often used to enhance

strategic clarity rather than to centralise authority (Saaty & Vargas, 2012b; Torfing, 2012).

Second, participation theory explains the normative dimension of governance legitimacy. High community involvement strengthens social acceptance, trust, and accountability (Nabatchi et al., 2017). However, participation alone does not guarantee effective coordination. As scholars of deliberative democracy argued, inclusive participation might increase voice diversity but could also produce decision-making complexity (Dryzek, 2022). Third, collaborative governance theory addresses the institutional mechanisms required to transform participation into coordinated action. Effective collaboration depends on facilitative leadership, trust-building, and formalised interaction arenas (Ansell & Torfing, 2022; Emerson & Nabatchi, 2015). Thus, participation and collaboration are related but not identical constructs.

More importantly, the expert-driven nature of AHP may appear to contradict participatory ideals. In this study, AHP was applied to key institutional actors representing different groups of stakeholders (directors, operational managers, and village supervisors), thereby embedding participatory representation within a structured decision framework. This approach reflects a hybrid governance model where technical rationality and social legitimacy coexist rather than compete. Therefore, the integrated framework does not assume that high participation automatically produces effective collaboration. Instead, it conceptualises governance as a layered system: AHP provides strategic prioritisation, participation secures legitimacy, and collaborative governance ensures operational coordination. This multi-level integration strengthens the analytical coherence of the proposed governance model.

2. Research Methods

This study aims to develop a sustainability-based VOE governance model that contributes to achieving the SDGs. Given the complexity of issues involving various dimensions of governance, this study employed the AHP method developed by Saaty (1980). AHP was chosen because of its ability to decompose complex decision problems into a systematic and measurable hierarchical structure as well as its capacity to combine qualitative and quantitative considerations in the assessment process (Saaty & Vargas, 2012a).

The population included all VOE with environmentally based business units in the study area. The sample was purposively determined by considering sustainability, accountability, and data completeness. Six VOE were selected as primary respondents, with each represented by a chief executive, operational manager, and treasurer familiar with the decision-making mechanisms and governance practices within their respective business units.

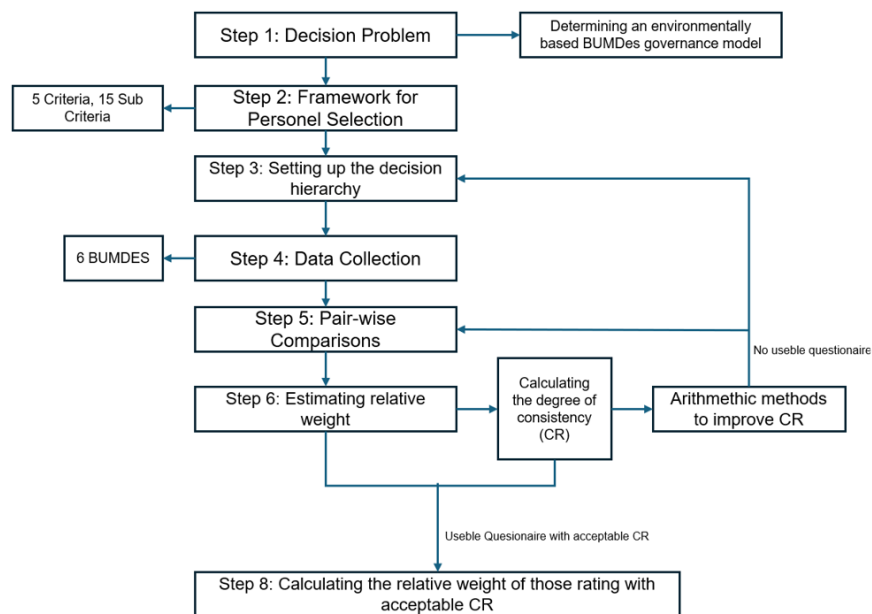


Figure 1. Research flow design using Analytic Hierarchy Process (AHP), adapted from Saaty (1980)

The AHP method was implemented in eight systematic steps, as shown in Figure 1. First, the research objectives were defined to determine the ideal sustainability-based VOE governance model that supported the achievement of the SDGs. Second, a hierarchical framework was developed consisting of 3 levels: (1) primary objectives; (2) 5 main criteria; and (3) 15 sub-criteria. The main criteria used in this study included: (a) human resource capacity and quality; (b) transparency and accountability; (c) collaboration and partnership; (d) environmental commitment; and (e) community participation and empowerment.

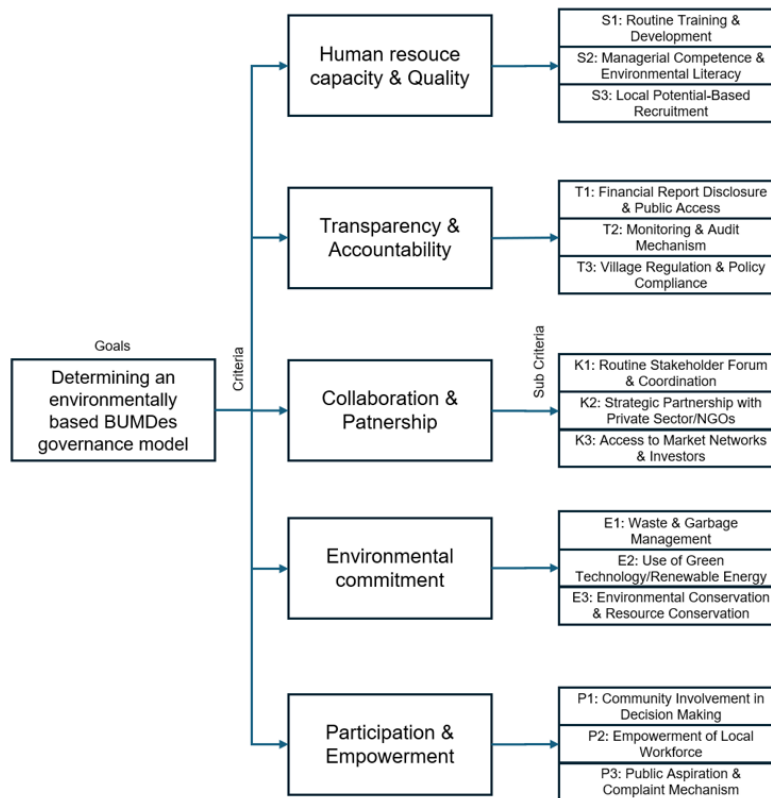


Figure 2. Criteria and subcriteria

The third step involved developing a decision hierarchy as shown in Figure 2, in which the relationships between the objectives, criteria, and sub-criteria are represented in a hierarchical structure. The fourth step, collecting weighting data, was conducted by distributing an AHP questionnaire to the six VOE managers. Each respondent was asked to provide a pairwise comparison assessment (pairwise comparison) between elements at a hierarchical level using Saaty's fundamental scale (1–9), where a value of 1 indicates the same level of importance. In contrast, a value of 9 indicates that one element is much more important than the other (Saaty, 2008).

The fifth step involved preparing a pairwise comparison matrix for each hierarchical level, both within and between the sub-criteria. Data from the questionnaire were compiled and averaged using the geometric mean to produce a combined comparison matrix. The sixth step, calculating the priority weights (priority vectors), involved determining the eigenvector values of each comparison matrix. These weights indicate the relative contribution of each element to the main objective.

The seventh step involved testing the consistency ratio (CR) to ensure that the respondents' assessments were logical and consistent. The CR value was calculated using the formula $CR = CI/RI$, where Consistency Index (CI) was obtained from $(\lambda_{\max} - n) / (n - 1)$, and the Random Index (RI) was determined based on the number of elements per matrix. Assessments were considered consistent if the CR value was ≤ 0.1 (Saaty & Vargas, 2012a). If $CR > 0.1$, the respondents were asked to review their assessments until an acceptable level of consistency was achieved.

The eighth step was global weight synthesis, which involved multiplying the weight of each subcriterion by the weight of its parent criterion to obtain the total priority for the entire hierarchical structure. The synthesis results were used to determine the priority order of the aspects of environmental-based VOE governance that most influenced sustainability and the achievement of the SDGs at the village level.

Through these stages, the AHP method not only produced a ranking of the criteria and sub-criteria that most significantly determined the effectiveness of sustainability-based VOE governance but also provided an analytical basis for developing a governance model that adapted to local conditions. The CR values for all the matrices in this study were below the threshold of 0.1; thus, the assessment results were considered consistent and accountable.

In addition to the AHP-based quantitative analysis, this study employed a qualitative approach through in-depth interviews to capture contextual insights into governance practices and sustainability orientation within VOEs. Semi-structured interviews were conducted with key informants from six selected VOEs, including directors, operational managers, and external parties from the village government, as they were directly involved in strategic decision-making and governance implementation. The qualitative data were analyzed using thematic analysis to systematically identify patterns, governance issues, and sustainability-related practices that informed the

development of the criteria and sub-criteria used in the AHP framework. This approach enabled the present study to bridge empirical narratives with structured decision analysis, thereby strengthening the analytical foundation of the governance model.

Thematic analysis followed a multi-stage coding process adapted from a previous study by Braun & Clarke (2006). First, all interview transcripts were read repeatedly to achieve data familiarization and to identify units with initial meanings related to governance, institutional capacity, environmental commitment, and community engagement. Second, open coding was applied to the data to inductively generate preliminary codes and then grouped them into broader themes aligned with five main governance criteria. To enhance analytical rigor, consistency of the coding was ensured through iterative comparisons across transcripts and reflexive memo writing, thus allowing the researchers to critically reflect on emerging interpretations. Validity was strengthened through data triangulation by comparing interview findings with documentary evidence and AHP results, while researcher bias was mitigated by maintaining an audit trail of coding decisions and systematically linking qualitative themes to quantifiable AHP criteria rather than subjective judgments. Through this process, qualitative insights served not only as descriptive support but also as an analytical foundation for structuring and validating the sustainability-based VOE governance model.

3. Results and Discussion

3.1 Profiles of Respondents

This study focused on six research objects, namely VOEs including Kemudo, Ponggok, Kenteng, Kopeng, Ngrawan, and Pager Sari. Each VOE has its own business unit, such as Ngrawan's water resource management and tourism village. Kenteng Village is a developing tourist village that manages environmental and community-based tourism. The potential of the village, including its beautiful natural surroundings, has developed into a natural tourist destination that attracts many tourists through the Sumber Arto Village-Owned Enterprise. Other potential opportunities managed by the Sumber Arto VOEs include the Agrotourism Market and Tahu Serasi. There are seven business units: Usaha Ekonomi Desa-Simpan Pinjam (UEDSP), Usaha Peningkatan Pendapatan Keluarga (UP2K), clean water management, sports, e-warung, waste bank management, and agrotourism market. Clean water management and waste bank business units have encountered several challenges, including inaccurate water discharge calculations, which result in incorrect water recording and payments, as well as a lack of separation of organic and inorganic household waste. For the management of natural tourism, clean trash bins and toilet facilities are required.

Ponggok Tourism Village, renowned for its thousands of springs, has received support from both village management and the government. The village management and government provide supporting facilities and infrastructure for tourist attractions, including swimming pools for adults and children, snorkeling and diving equipment rentals, the addition of freshwater fish species, amusement rides, the addition of various types of beautiful water pools, and the maintenance of pristine water sources. Furthermore, the construction of physical support infrastructure encompasses restaurants/food stalls, parking lots on community land, gazebos, restrooms, prayer rooms, grocery stores, tourist markets, and tourist shops. Emerging issues include the threat of climate change, such as fluctuations in rainfall and the risk of drought, as well as the declining quality of the micro-watersheds surrounding Ponggok Village, which have the potential to impact water supply. Preliminary studies have indicated that a 15% decrease in vegetation along the village's riverbanks over the past decade has directly impacted the flow and clarity of the springs. Without immediate conservation efforts, the carrying capacity of water-based tourism would become increasingly vulnerable. Furthermore, social inclusivity is also an issue; not all community groups, particularly women, youth, and vulnerable groups, are directly involved in tourism economic activities. Without participatory governance, social inequality could widen and become more widespread.

Furthermore, the Tirta Umbul Rejo VOE of Kopeng Village manages several tourist destinations, such as the Kopeng Tourist Market. The Kopeng Tourist Market spans an area of approximately 29,400 m² and is part of the tourist attractions managed by the Kopeng VOE. It is a popular stopover for many visitors to Kopeng Village, where they shop for fresh vegetables, fruits, and ornamental plants. Currently, the Kopeng Tourist Market serves as the key tourist attraction in Kopeng Village, with potential for further tourism development. Problems that arise from the functional side of the market include a lack of cleanliness, which is caused by inadequate spatial planning facilities. From an architectural and climate change perspective, rain and fog during dry and rainy seasons disrupt visibility and shopping comfort. Some stalls were bent, thus causing the floor to become muddy.

In this study, sustainability is defined as the institutional capacity of VOEs to maintain long-term economic viability, while simultaneously safeguarding environmental resources and promoting inclusive social benefits within the village context. Sustainability is not treated merely as an environmental outcome or financial performance indicator but as a governance-driven process embedded in decision-making structures, operational practices, and stakeholder engagement mechanisms. This definition aligns with sustainability governance

perspectives that emphasize the integration of economic, social, and environmental dimensions (triple bottom line) and views local institutions as key actors in translating SDGs into context-specific practices (Bebbington & Unerman, 2018; Biermann et al., 2017; Elkington, 1997). Accordingly, sustainability in this research was operationalized through governance criteria that reflected the capacity of VOEs to manage human resources, ensure transparency and accountability, build collaborative networks, uphold environmental commitments, and empower local communities.

The six VOEs examined in this study represent diverse typologies of VOEs based on their dominant business orientation and sustainability challenges. Ngrawan and Ponggok are categorized as natural resource-based and water-dependent enterprises, where sustainability is closely linked to the conservation of springs, water quality, and ecosystem carrying capacity amid climate change. Kenteng and Kopeng are environmental tourism-based enterprises that rely on landscape quality, waste management, sanitation infrastructure, and community-based tourism services to sustain visitors' demand while minimizing ecological degradation. Meanwhile, Pager Sari and Kemudo VOEs represent mixed-type enterprises that combine the provision of public services, market management, waste banking, clean water services, and agrotourism activities. This categorization is analytically important, as each type faces distinct governance and sustainability risks ranging from resource depletion and climate vulnerability to waste management inefficiencies and social inclusion gaps. By explicitly acknowledging these variations, this study strengthened the analytical coherence of the sustainability-based governance model and enhanced its relevance in understanding how different forms of VOEs contribute to SDGs implementation at the local level.

3.2 Thematic Analysis of Governance and Environmental Orientation

3.2.1 From volunteer to professional: The transformation of Village-Owned Enterprises' management capacity

Human resource (HR) capacity emerged as the most dominant and recurring theme throughout the interviews. VOE managers have revealed that managerial and technical skills remain key challenges in managing environmental business units.

The Ngrawan VOE manager explained that their biggest obstacles “lie in limited human resource management and suboptimal promotional strategies”, preventing significant growth in business units outside the spring. A similar sentiment was echoed by the Tirta Umbul Rejo (Kopeng) VOEs, which emphasized that “each division requires personnel with strong expertise in their field, but finding such qualified personnel is still difficult in the village”.

Meanwhile, the Kenteng VOEs face a structural dilemma: “Its core management is not even paid a fixed salary... because the business's performance has not been optimal”, said the Kenteng Village Secretary. This situation suggests that management motivation and retention are significant factors that affect the continuity of operational activities. This issue of human resource capacity concerns not only technical skills, but also professionalism and motivation. Some managers rely on volunteer work systems that are not always accompanied by ongoing training. Consequently, VOE management often relies on individual initiatives rather than a well-established institutional system.

This finding aligns with Said et al. (2021), who found that the performance of VOEs in Indonesia was significantly influenced by the quality of their human resources, particularly managerial skills, financial record keeping, and product innovation. Therefore, strengthening human resources is urgently required for the transformation of VOEs from social institutions into sustainable economic entities.

Furthermore, the interviews indicated that VOEs with a planned training system tended to have higher productivity. This is evident in Ponggok VOEs, which have developed a water tourism business unit staffed by professional divers and photographers. According to one of its administrators, “We once sent young people to dive training in Yogyakarta... so they could independently photograph guests and sell the photos”.

The Ponggok case highlights the importance of planned capacity building in fostering innovation and professionalism, a practice rarely observed in other VOEs that often operate primarily with volunteers. However, adequate human resource capacity and quality will not yield optimal results without transparent and accountable governance. As emphasized by the managers of VOEs, Ngrawan and Kenteng, public trust in VOEs is strongly influenced by transparency in fund management, reporting, and business results. Therefore, the next dimension that needs to be considered is transparency and accountability, as the foundation of institutional integrity.

3.2.2 Building trust from below: Transparency as social legitimacy

Transparency and accountability emerged as a crucial dimension in the interviews. Several managers emphasized the demand for financial transparency as a means of building public trust. However, the implementation remains inconsistent across VOEs. According to the administrator of Kenteng VOEs, accountability meetings and village deliberations are held only “twice a year”. At the same time, financial reporting “cannot be disclosed routinely because it is still compiled manually”. A similar situation exists in the Ngrawan VOEs, where the village government has allocated a budget to support business units, but “budget absorption has

not been optimal” due to weak coordination between administrators and village officials.

Some VOEs are facing challenges in managing their internal audit systems. Tirto Umbul Rejo, a VOE administrator, stated that “report transparency is important, but we do not have dedicated administrative staff, so everything is handled by the administrators, who also handle operational responsibilities”. The lack of an adequate administrative structure makes accountability an additional burden for administrators rather than a systematically functioning institutional mechanism. This highlights the disparity between modern governance principles and the technical capabilities of village institutions. As explained by Sulistyorini et al. (2024), public accountability in VOEs could only be upheld if accompanied by a transparent, simple, and publicly accessible reporting mechanism.

Furthermore, Pratiwi et al. (2024) emphasized that financial transparency was not only an instrument of social control but also a factor in strengthening institutional legitimacy, capable of influencing the level of citizens’ participation in village economic activities. In this context, transparency is not merely an administrative obligation but also a strategy for building social trust (trust-based governance). Several VOEs have attempted to implement simple digital approaches, such as using social media to communicate business activities. However, these efforts have not been followed by financial digitization. This suggests an opportunity to develop a technology-based reporting system that is more adaptive to local capacity, for example, through a publicly accessible village reporting dashboard.

Strong transparency and accountability practices enhance the social legitimacy of VOEs in the eyes of the public; however, the sustainability of these institutions is also largely determined by their ability to establish and maintain relationships with external parties. As stated by the administrators in Ponggok and Kopeng VOEs, capital and market constraints are often overcome through cross-sector collaborations. Therefore, collaboration and partnerships are the next strategic steps in optimizing governance and expanding the socioeconomic impact of VOEs.

3.2.3 Villages’ social and economic networks: Collaboration as collective capital

Community collaboration is a crucial pillar of environment-oriented VOE governance. Nearly all respondents acknowledged that community support was a distinguishing factor between thriving VOEs and stagnant VOEs. In Ngrawan VOEs, the roles of local communities, such as Karang Taruna and Omah Cikal, are particularly prominent. “The young people organize music events and festivals, so VOEs only help with the facilities,” said a local administrator. This community involvement extends beyond labor and provides ideas and innovations for activities that have both economic and social value. Furthermore, Ngrawan Village collaborates with local villages with similar businesses, particularly tourism, to promote each other and combine tour packages for specific events. Meanwhile, Ponggok VOEs collaborate with private partners, such as Bank Negara Indonesia (BNI) and tourism communities, which assist in the development of facilities and the promotion of water destinations.

However, not all VOEs exhibit similar participation dynamics. Pagersari VOEs revealed that previous VOE management failures caused “communities to be reluctant to participate,” leading them to focus on rebuilding public trust through social activities and training. This demonstrates that community participation is dynamic and influenced by the history of social interactions within the village. As Kakumba (2010) noted, the sustainability of village governance was determined not only by formal structures but also by “the level of social trust and the quality of horizontal communication among residents”.

In the context of environmental-based VOEs, community participation in nature conservation activities is also evident. For example, the Kenteng VOEs plan to develop a waste bank and a 3R waste disposal site (TPS 3R), but face challenges of participation because the community is “not yet accustomed to sorting waste”. It emphasizes that participation is not merely about engaging in deliberations, but rather about collective behavioral change that requires education and social leadership from VOEs.

On the other hand, efforts to build external networks and business partnerships should be accompanied by responsibility for the sustainability of the environment in which VOEs operate. Strong collaboration aims not only to improve economic performance but also to create mechanisms to protect the natural resources that form the primary basis of the village economy. Therefore, the following discussion highlights the environmental commitment dimension, which is at the core of sustainability-based VOE governance.

3.2.4 Ecology as business ethics: Negotiating profit and nature

Commitment to environmental sustainability is explicitly evident in several VOEs that manage natural resources, particularly water, waste, and tourism. The Tirto Umbul Rejo (Kopeng) VOEs and Ponggok VOEs are the most obvious examples. In Copenhagen, the management stated that “water sources need to be protected because the flow decreases during the dry season”, leading them to implement a water usage restriction system for specific business units.

In Ponggok, environmental conservation is an integral part of the business strategy. Management explained, “We regularly plant trees in the upstream areas of Merapi to stabilize the water flow, because if the upstream is damaged, our business will also be impacted”. This approach reflects strong ecological awareness, where economic sustainability is understood as a direct function of resource conservation and preservation. However,

environmental commitment is not without its financial or bureaucratic challenges. In Ponggok for example, management complained about the “50 million rupiah per year water rental tax burden”, which they considered hindered the development of environmental and social programs. Meanwhile, Kenteng VOEs face limited funds to maintain the 3R waste disposal site (TPS 3R) and educate the community about waste management.

Both cases illustrate the dilemma between economic and ecological goals: Preserving the environment requires costs, whereas the fiscal capacity of VOEs remains constrained. These findings support the view that limited financial and institutional resources often hamper the integration of environmentally friendly practices at the village level (Lazos-Ruíz et al., 2014). Therefore, external support in the form of green incentives, Corporate Social Responsibility (CSR) partnerships, or revolving environmental funds is necessary to enable VOEs to perform their ecological functions without compromising their financial sustainability.

These efforts demonstrate that environmental sustainability at the village level is inextricably linked to the social and cultural capacities that underlie it. Ecological awareness is built not only through regulations or technical interventions but also through the internalization of collective community values. In the context of VOEs, a strong environmental commitment necessitates social support, specifically the participation of residents who understand, accept, and actively engage in efforts to conserve the village’s natural resources. Therefore, the dimensions of community participation and empowerment are integral pillars of successful environmentally-based governance of VOEs. This aspect not only strengthens the institution’s social legitimacy but also ensures that any environmental policies or programs have strong social roots at the community level.

3.2.5 From spectator to driver: Enlivening citizen participation in Village-Owned Enterprises

Community participation and empowerment have emerged as important dimensions that strengthen the governance of sustainability-based VOEs. The interviews indicated that the success of VOEs in developing sustainable business units was significantly influenced by the active involvement of residents in planning, implementing, and monitoring activities. In Ngrawan VOEs, community participation is evident through the support of local communities such as Karang Taruna and Omah Cikal, which independently organized arts activities and promoted village tourism. One administrator explained, “Young people are now doing a lot of the work; we at VOEs just help with the facilities and costs”. This pattern demonstrates a form of empowerment based on community initiatives, in which VOEs act as facilitators rather than simply controlling village economic activities.

In contrast, Pagersari VOEs face the challenge of low community participation due to negative experiences during the previous management period. One administrator stated that “the community initially lacked trust because of previously failed VOEs, so now we are starting from scratch to rebuild trust”. This case emphasizes that community participation is inextricably linked to trust and the history of social interaction within the village. The empowerment process must begin with the reconstruction of social relationships that previous institutional experiences have damaged.

Meanwhile, at Kenteng VOEs, community participation began to focus on environmental issues through a 3R waste management plan and waste bank. However, the management acknowledged that they still faced behavioral resistance because “people are not yet accustomed to sorting waste”. This indicates that empowerment involves not only physical involvement but also a transformation of awareness and collective behavioral change, which requires time and effective communication strategies.

The above findings align with the participation theory of Arnstein (1969), which places citizen involvement on a ladder of participation, ranging from informative to collective decision-making (citizen power). In the context of VOEs, ideal participation extends beyond symbolic attendance at village meetings to encompass learning and collective decision-making. Petty & Krosnick (2014) explained that transformative participation increased community ownership of the program, strengthened social solidarity, and ensured the sustainability of economic and environmental activities.

Beyond participation, community empowerment encompasses local workforce involvement and the development of entrepreneurial capacity. Several VOEs, such as Ponggok and Kopeng have successfully created jobs for the surrounding community through water tourism and clean water management businesses. One Ponggok administrator stated that “every holiday season, we can recruit up to 50 residents as cleaners and parking attendants”. These activities not only improve economic well-being but also strengthen VOEs’ social legitimacy as inclusive public institutions.

However, community economic empowerment still has limitations in terms of access to training and business capital. Some residents lack the entrepreneurial capacity to fully capitalize on the opportunities generated by VOEs. This reinforces the argument of Chambers (1997) that true empowerment requires redistribution of economic power through increased access to resources, training, and opportunities to participate in the production process.

Thus, community participation and empowerment are at the heart of VOE governance that prioritizes social sustainability. Without active community support, VOEs risk becoming merely administrative entities, without a strong social foundation. Therefore, VOEs’ development strategies should position empowerment as a long-term process, starting with building trust, strengthening community socioeconomic capacity, and establishing fair and

sustainable participation mechanisms.

As emphasized by Kakumba (2010), the success of village governance depends not only on the effectiveness of formal structures but also on the strength of social capital and citizen participation as key actors in the development process. By integrating participation into institutional design, VOE can serve a dual function: as a driver of the village economy and as a collective learning space for sustainable development.

3.2.6 Analytic Hierarchy Process-based governance criteria

Quantitative analysis using the AHP, as shown in Table 1, yielded a priority ranking of sustainability-based VOE governance that reinforced qualitative findings from in-depth interviews. The CR value = $0.06 < 0.1$, indicating that perceptions between respondents were consistent and reliable (Saaty, 2008). The AHP results showed that the Human Resource Capacity and Quality dimension (0.3333) ranked first, followed by Transparency and Accountability (0.2667), Collaboration and Partnership (0.2000), Environmental Commitment (0.1333), and Community Participation and Empowerment (0.0667).

Table 1. Criteria weights

Criteria	Results	Priority
Human Resources Capacity & Quality	0.333333	1
Transparency & Accountability	0.266667	2
Collaboration & Partnership	0.200000	3
Environmental Commitment	0.133333	4
Community Participation & Empowerment	0.066667	5

3.2.7 Human resource capacity and quality

This criterion ranked highest, as shown in Table 2, indicating that human capacity is the primary foundation for effective VOE governance. The highest sub-criteria were Managerial Competence and Environmental Literacy (0.1667), followed by Routine Training and Development (0.1111), and Local Potential-Based Recruitment (0.0556). The weighting calculation for the scope of the HR capacity and quality criteria is also consistent with the final weighting results, with Managerial Competence and Literacy having a weighting of 0.5, Routine Training and Development having a weighting of 0.333, and Local Potential-Based Recruitment having a weighting of 0.1667.

Table 2. Sub-criteria weights

Criteria	Sub-Criteria	Relative Weight of the Five Criteria Matrices	Final Weight
Human Resources Capacity & Quality	S1: Routine Training & Development	0.333333	0.111111
	S2: Managerial Competence & Environmental Literacy	0.500000	0.166667
	S3: Local Potential-Based Recruitment	0.166667	0.055556
	T1: Financial Report Disclosure & Public Access	0.500000	0.133333
Transparency & Accountability	T2: Monitoring & Audit Mechanism	0.333333	0.088888
	T3: Village Regulation & Policy Compliance	0.166667	0.044445
	K1: Routine Stakeholder Forum & Coordination	0.333333	0.066666
Collaboration & Partnership	K2: Strategic Partnership with Private Sector/NGOs	0.500000	0.100000
	K3: Access to Market Networks & Investors	0.166667	0.033333
	E1: Waste & Garbage Management	0.500000	0.066667
Environmental Commitment	E2: Use of Green Technology/Renewable Energy	0.166667	0.022222
	E3: Environmental Conservation & Resource Conservation	0.333333	0.044444
	P1: Community Involvement in Decision Making	0.333333	0.022222
Participation & Empowerment	P2: Empowerment of Local Workforce	0.500000	0.033333
	P3: Public Aspiration & Complaint Mechanism	0.166667	0.011111

This finding confirms that increasing human capacity, both technical and conceptual, is key to the success of

village economic institutions (United Nations Development Programme, 2009). Environmental literacy has become a new aspect that is strengthening in the era of village green transition, in line with the findings of Lazos-Ruiz et al. (2014), that community-based economic institutions that integrated ecological knowledge into business practices were more adaptive to socio-environmental risks.

In this context, human resources are viewed not only as program implementers but also as transformational actors shaping institutional values (Velazquez et al., 2011). Continuous training enables social innovation, whereas recruitment based on local potential strengthens the VOs' social legitimacy in the eyes of residents.

3.2.8 Transparency and accountability

Transparency and accountability emerged as the second priority criteria, demonstrating that the social legitimacy of VOs depends on their administrative integrity and transparency. The primary sub-criteria were Financial Report Transparency and Public Access (0.1333), followed by Monitoring and Audit Mechanisms (0.0889) and Compliance with Village Regulations and Policies (0.0444). Practices of open public accountability play a crucial role in strengthening public trust in institutions (Bovens, 2007). The interview results indicated that VOs that implemented open and easily accessible financial reports had higher levels of participation and support. Transparency also serves as a social control mechanism that ensures that decision-making is based on the principles of distributive justice (Kaufmann & Kraay, 2008). In the village context, transparency is not only administrative but also cultural, forming trusting social relationships and minimizing the potential for internal conflict.

3.2.9 Collaboration and partnership

The third criterion highlights the role of networks and external relationships in strengthening the VOs' capacity. The sub-criterion with the highest weighting is strategic partnerships with private sectors/NGOs (0.1000), followed by Routine Stakeholder Forums and Coordination (0.0667) and Access to Market Networks and Investors (0.0333).

As emphasized by Ostrom (1990) and Lockwood (2010), healthy partnerships between community institutions and external actors enhance institutional resilience. In the context of VOs, partnerships encompass access to capital, knowledge transfer, managerial innovation, and marketing networks. Effective partnerships require regular and planned coordination across sectors. Without a structured collaboration system, collaborative initiatives often lack structure and tend to be personal and short-term in nature. Therefore, VOs must develop a stakeholder engagement model based on mutually beneficial partnerships to strengthen business sustainability.

3.2.10 Environmental commitment

Environmental commitment ranked fourth but had high thematic significance in the context of natural resource-based VOs. The Waste and Garbage Management sub-criteria (0.0667) and Environmentally-Friendly Innovation and Energy Efficiency (0.0444) stood out as top priorities, followed by Ecological Education and Awareness (0.0222).

The integration of ecological values into VO governance demonstrates a trend toward local-level eco-governance (Lazos-Ruiz et al., 2014). Although less important than managerial aspects, environmental commitment serves as an indicator of an institution's moral responsibility for long-term sustainability. As conveyed by the management of Kenteng VOs, the main challenge is changes in community behavior towards waste management, implying that technical innovation must be balanced with social and educational strategies.

3.2.11 Community participation and empowerment

This criterion carries the smallest weight, but serves as the moral and social pillar of the overall governance system. The sub-criteria of Community Involvement in Strategic Decisions (0.0267) and Social Inclusion and Representation (0.0267) had nearly equal weights, demonstrating the importance of democratizing decision-making at the village level.

This finding aligns with Arnstein (1969), who placed active citizen participation as the highest form on the participation ladder, and Petty & Krosnick (2014), who emphasized transformative participation as a means of social empowerment.

Despite its relatively minor role, community participation serves as a social bond between the actors in village governance. Without participation, governance practices risk becoming top-down and losing the collective support that is necessary for sustainability.

3.2.12 Comparative governance patterns across Village-Owned Enterprise types

Beyond categorising the six VOs into natural resource-dependent (Ngrawan and Ponggok), eco-tourism oriented (Kenteng and Kopeng), and hybrid models (Pager Sari and Kemudo), this study further examined how governance dynamics varied across these types. Sustainability in this research was conceptualised through the triple bottom line framework, integrating economic continuity, environmental stewardship, and inclusive social value creation (Hahn et al., 2015; Schaltegger et al., 2019). VOs highly dependent on natural resources demonstrate not only stronger ecological sensitivity but also greater vulnerability to risks of resource dependency.

As suggested by adaptive governance literature (Djenontin & Meadow, 2018; Haller et al., 2016), organisations managing common-pool resources tend to exhibit structures with tighter control due to ecological risk exposure. In Ponggok and Ngrawan, environmental commitment is institutionally embedded, yet decision-making authority remains relatively concentrated among key actors to ensure protection of resources. This reflects patterns identified in commons governance studies, where ecological vulnerability shapes governance centralisation.

In contrast, eco-tourism-oriented VOEs such as Kenteng and Kopeng show broader community engagement but face challenges of operational coordination (see Table 3). Community enterprise literature (Ohe, 2020; Steiner & Teasdale, 2019) suggested that social enterprises operating in tourism ecosystems relied more heavily on participatory legitimacy, which might explain relatively higher emphasis on collaboration and transparency within these cases. Hybrid VOEs (Pager Sari and Kemudo) demonstrate more complex governance arrangements due to diversified revenue streams. Such hybridity aligns with institutional logics theory, in which organisations must balance economic and social logics simultaneously (George et al., 2016). This complexity is reflected in AHP results showing more balanced weighting across governance criteria.

Table 3. Comparative governance characteristics across Village-Owned Enterprise (VOE) types

Governance Characteristic	Natural Resource-Dependent VOEs (Ngrawan, Ponggok)	Eco-Tourism Oriented VOEs (Kenteng, Kopeng)	Hybrid VOEs (Pager Sari, Kemudo)
Resource Dependence Pattern	High dependence on water resources and ecological assets (springs and river ecosystems).	Dependent on tourism flows and environmental attractiveness.	Diversified income streams (economic services + environmental/social units).
Decision-Making Structure	Relatively centralized to safeguard ecological assets and maintain resource control.	Semi-participatory; operational decisions involve multiple local stakeholders.	More complex governance arrangements balancing economic and social functions.
Community Embeddedness	Moderate-High (economic participation present but ecological management concentrated among core actors).	High community interaction, but uneven inclusion of vulnerable groups.	Moderate-High; broader institutional linkage with village government and community groups.
Operational Sustainability Risks	Climate variability, water debit fluctuation, and ecosystem degradation.	Service quality decline, waste management issues, and infrastructural limitations.	Coordination complexity, managerial overload, mission drift risk.
Dominant Governance Emphasis (AHP Interpretation)	Although environmental commitment was formally weighted lower in AHP, field evidence indicated that ecological stewardship acted as a structural constraint underlying all governance decisions. HR capacity prioritized for management of technical resources.	Transparency and collaboration received relatively stronger emphasis due to the demand for the coordination of the tourism ecosystem. HR capacity was important for service management and visitor satisfaction.	AHP results showed a more balanced distribution across criteria, thus reflecting institutional hybridity. HR capacity remained dominant, but collaboration gained strategic importance.

Table 4. Thematic-based interpretation of Analytic Hierarchy Process (AHP) criteria across Village-Owned Enterprise (VOE) types

Governance Criteria	AHP Weight	Natural Resource-Dependent VOEs (Ngrawan, Ponggok)	Eco-Tourism VOEs (Kenteng, Kopeng)	Hybrid VOEs (Pager Sari, Kemudo)
Human Resource Capacity	0.3333	High	High	High
Transparency & Accountability	0.2667	Medium	High	High
Collaboration & Partnership	0.2000	Medium	High	High
Environmental Commitment	0.1333	Very High	Medium	High
Community Participation	0.0667	Medium	High	Medium

More importantly, while AHP results indicated human resource capacity and transparency as dominant priorities overall, comparative observation suggested that governance prioritisation was partially shaped by VOE typology. Managers in natural resource-dependent VOEs exhibit stronger implicit prioritisation of environmental safeguards, even when weighted scores appear lower. This finding supports the sustainability paradox argument (Bansal & Song, 2017), in which strategic decision-makers may understate ecological variables in formal ranking yet operationally treat them as foundational constraints. Thus, integrating typology with AHP results, as shown in Table 4, enriches interpretation by revealing that governance priorities are not uniform but contextually embedded

within resource dependence patterns, community structure, and institutional hybridity.

This study applied an interpretative quantification approach by translating qualitative thematic findings into relative intensity categories (very high, high, medium, and low). This approach is consistent with mixed-methods research, where qualitative insights are systematically transformed into comparative analytical categories to complement quantitative results (Creswell & Clark, 2017; Guest et al., 2011). Rather than recalculating AHP weights for each VOE type, which is not methodologically feasible due to the aggregated nature of pairwise comparisons, this approach enables contextual interpretation of global priorities across different governance settings.

The classification of intensity levels is based on three criteria: (1) the frequency of theme occurrence across interview data; (2) the emphasis placed by respondents; and (3) the functional role of each governance dimension in operational decision making. This procedure aligns with thematic coding and pattern identification approaches in qualitative analysis, in which meaning is constructed through repeated patterns and their practical significance (Saldaña, 2021). A governance criterion is categorized as high when it is consistently mentioned, strongly emphasized, and directly influences decision making; medium when it is relevant but not dominant; and low when it appears primarily as a normative or peripheral element. The very high category is assigned when a criterion functions as a structural constraint shaping all governance decisions.

The findings indicated that while human resource capacity remained consistently dominant across all VOE types, other governance dimensions exhibited context-dependent variation. For example, environmental commitment was interpreted as very high in natural resource-dependent VOEs, despite its lower AHP weight (0.1333), as it acted as a binding ecological constraint. This reflects the sustainability paradox, in which environmental considerations may be underrepresented in formal prioritization yet remain critical in practice (Bansal & Song, 2017). In contrast, eco-tourism VOEs demonstrate stronger emphasis on transparency and collaboration due to coordination demands in service-based ecosystems, while hybrid VOEs exhibit a more balanced governance configuration to reflect their institutional complexity.

4. Discussion

The integration of thematic analysis and the AHP demonstrated consistency and complementarity in formulating an environment-based VOEs governance model that supported the achievement of SDGs as shown in Table 5. In-depth interviews emphasized the importance of human resource capacity, transparency, collaboration, environmental commitment, and community participation as key factors for sustainability. Meanwhile, quantitative analysis using AHP validated this priority order (Cheng & Li, 2001; Khorramshahgol & Al-Husain, 2021), placing human resource capacity and quality at the top, followed by transparency and accountability, collaboration and partnerships, environmental commitment, and community participation.

Table 5. The relationship between the sustainability-based Village-Owned Enterprise (VOE) governance model and the Sustainable Development Goals (SDGs)

Analytic Hierarchy Process (AHP) Criteria	Relevance to SDGs	Explanation of the Relationship
Human Resources Capacity & Quality	SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth)	Improving managerial competencies, environmental literacy, and regular training reflects lifelong learning (SDG 4), while strengthening locally based productive employment opportunities (SDG 8).
Transparency & Accountability	SDG 16 (Strong, Peaceful and Inclusive Institutions)	Open governance, transparent financial reporting, and public audit mechanisms strengthen social legitimacy and create accountable and trustworthy village institutions.
Collaboration & Partnership	SDG 17 (Partnerships to Achieve the Goals), SDG 8 (Decent Work and Economic Growth)	Cross-sector collaboration among the private sector, NGOs, and communities expands access to markets, capital, and innovation, thus driving village economic growth and strengthening development networks.
Environmental Commitment	SDG 12 (Responsible Consumption and Production), SDG 13 (Addressing Climate Change), SDG 15 (Terrestrial Ecosystems)	Environmentally friendly practices, such as water conservation, waste management, and the use of green technologies, support sustainable consumption and production, climate change mitigation, and the preservation of village natural resources.
Participation & Empowerment	SDG 11 (Sustainable Cities and Human Settlements), SDG 5 (Gender Equality), SDG 10 (Reduced Inequalities)	Active community involvement in decision making and the empowerment of the local workforce create inclusive villages, reduce social disparities, and support community-based and sustainable settlements.

Although the environmental commitment aspect ranked fourth in the AHP results, interviews with VOE

managers indicated that this aspect played a significant role in field practice. Concrete examples are seen in Ponggok VOs, which integrates upstream water conservation and reforestation into its business strategy, and Tirta Umbul Rejo VOs, which implements water usage restrictions during the dry season to maintain resource sustainability. These practices demonstrated that village economic sustainability could not be separated from ecological sustainability. Therefore, the environment should be positioned as the primary basis for VO governance. This perspective aligns with the views (Dolšák & Ostrom, 2003; Ostrom, 1990; Ostrom, 2015) regarding the importance of joint management of natural resources (commons), while also strengthening the relevance of VOs to SDG 12 on responsible consumption and production and SDG 13 in addressing climate change.

Further, of its relevance to environmental aspects, sustainability-based VO governance directly contributes to SDG 4. Research indicated that enhancing human resource capacity through training, mentoring, and developing ecological literacy not only strengthened managerial performance, but also served as a tool for non-formal education at the village level (Brandi & Thomassen, 2021; Feeney et al., 2023). Thus, VOs play a role in expanding community access to lifelong learning, which aligns with the SDG 4 agenda regarding quality education (Rasheed et al., 2025; Razali & Jamil, 2023).

The research also showed that human resource capacity was not only related to managerial skills but also ecological literacy, which enabled village managers to be more adaptive in facing socio-environmental risks. Ecological literacy is a crucial asset for the development of environment-based business units, such as water management, waste management, and ecotourism. Thus, human resources play a role not only as technical implementers, but also as transformational actors capable of connecting economic, social, and environmental interests (Velazquez et al., 2011). This finding aligns with the capacity-building framework outlined by the United Nations Development Programme (2009), which stated that improving human capacity was a fundamental pillar for realizing sustainable development.

Inclusive governance encourages job creation based on local potential in tourism businesses, clean water management, and waste management, thereby improving community welfare and supporting the achievement of SDG 8 on decent work and economic growth (Arifin et al., 2020; Badaruddin et al., 2021). In other words, VOs are not only economic instruments but also social development agents that strengthen human capacity and open productive employment opportunities in rural communities (Aritenang, 2021; Imanuella et al., 2025).

On the other hand, transparency and accountability, which ranked second in the AHP results, are also important foundations for successful environmental governance. VOs that implement open and publicly accessible financial reports tend to be trusted more by the community to manage village resources. This kind of transparency serves not only as an administrative instrument (Ponce, 2005), but also as a social legitimacy mechanism that strengthens citizen participation (Bovens, 2007; Mashaw, 2006; Willems & Van Dooren, 2012). With strong legitimacy, environmental programs such as waste management or water conservation are more easily supported collectively, thereby strengthening VOs' contribution to achieving SDG 16, which concerns the establishment of strong, peaceful, and inclusive institutions.

Another prominent finding is the importance of cross-stakeholder collaboration as a mechanism for supporting sustainability. Interviews revealed that VOs that partnered with the private sector, NGOs, and local communities could overcome the limitations in capital, technical skills, and market networks. This type of collaboration reflects the mutually beneficial nature of partnerships, as emphasized by Lockwood (2010), while also strengthening the relevance of VOs to SDG 17, which focuses on partnerships to achieve development goals.

Community participation, with the smallest weighting in the AHP results, remains a crucial social pillar that ensures the ecological sustainability of the village. Without collective behavioral changes, such as the habit of sorting waste or preserving water, the environmental efforts of VOs will not be sustainable. This reinforces the theory of participation (Arnstein, 1969), which states that active citizen involvement is the highest form of participation and a prerequisite for successful inclusive development. In this context, community ecological participation is closely linked to SDG 11 in sustainable settlements.

The synthesis of these thematic and quantitative results demonstrates that sustainability-based VO governance that supports the SDGs should be developed in an integrative manner. Ecologically minded human resource capacity, transparency that fosters social legitimacy, multi-stakeholder collaboration that expands technical and financial capacity, and community participation as a socio-ecological foundation are the complementary pillars. Without a combination of these four pillars, risk derived from VO is partial as it loses impetus toward sustainability. Thus, VOs can be viewed not only as a driver of the village economy, but also as an agent of conservation and social transformation that contributes directly to achieving the global sustainable development agenda.

5. Implications and Contributions of the Research

The results of this study have significant theoretical and practical implications for the development of sustainability-based VO governance within the framework of achieving SDGs. Based on the results of the AHP,

the criteria of human resource capacity and quality (HR) occupied the highest weight of 0.3333, followed by transparency and accountability (0.2667), collaboration and partnership (0.2000), environmental commitment (0.1333), and community participation and empowerment (0.0667). This weight configuration suggests that the sustainability of VOE is not only determined by technocratic capabilities but also by institutional capabilities to create cooperation among humans, governance, and environmental factors.

Theoretically, this study expanded the village governance model toward an eco-governance paradigm in which human, institutional, and ecological dimensions were intertwined. These results confirmed the findings of Velazquez et al. (2011) that sustainable learning organizations were supported by the human capacity to internalize environmental values into institutional practices. Furthermore, the dominance of human resources in the AHP priority structure reinforced the thesis of the United Nations Development Programme (2009), that human capacity development was a core driver in building resilient local institutions that were adaptive to changes in the social and ecological environment. Thus, the conceptual contribution of this study lies in the integration of human-centered governance with environmental stewardship, a concept that has rarely been addressed in previous studies on VOE governance (Kania et al., 2021; Sulistyorini et al., 2024).

Despite their low weighting, environmental commitment and community participation reinforced Ostrom (1990)'s notion of commons governance, namely the importance of natural resource management based on social collaboration and local norms. This study confirmed that ecological literacy and collective participation served as moral bonds in villages' economic sustainability. This finding extended Arnstein (1969)'s theory of participation from a political context to social ecology.

The apparent discrepancy between the low AHP weight assigned to community participation (0.0667) and its position as the "highest form" in participation theory could be reconciled by distinguishing between normative ideals and operational priorities. Participation theory, particularly Arnstein's ladder, conceptualizes participation as the ultimate stage of governance when citizens gain meaningful control over decision-making (Arnstein, 1969). However, this normative ideal does not imply that participation should always emerge as the immediate priority in all institutional contexts. Instead, the findings of this study reflect a phased governance dynamic, in which institutional readiness, especially human resource capacity and governance structure, serves as a prerequisite for effective participation. Previous studies highlighted that participatory processes required adequate institutional capacity, trust, and facilitative structures to function meaningfully, and without these conditions, participation might become symbolic or ineffective (Akamani & Hall, 2015). Moreover, participation does not automatically translate into effective collaboration, as collaborative governance depends on structured coordination mechanisms and shared institutional arrangements beyond mere inclusion (Barrutia & Echebarria, 2019). Therefore, the low AHP weighting does not indicate a diminished importance of participation, but rather reflects its role as a long-term ethical and institutional objective. Its effectiveness depends on prior development of governance capacity. In this sense, technocratic prioritization and participatory ideals should be viewed as complementary stages within a progressive governance framework rather than contradictory elements.

Beyond its relevance to VOE in Indonesia, the implications of this study extend to broader discussions of governance mechanisms for community-based and locally owned enterprises in developing and emerging economies. Although VOE is institutionally embedded within the Indonesian village governance system, the identified governance challenges, such as limited human capacity, weak accountability mechanisms, fragmented collaboration, and environmental pressures, are not unique to Indonesia. Similar organizational forms, including community enterprises, cooperatives, and social enterprises, operate as local economic actors in many countries across Asia, Africa, and Latin America, where they serve as intermediaries between grassroots development initiatives and national sustainability agendas (Doherty et al., 2014). In this regard, the sustainability-based VOE governance model proposed in this study offers a transferable analytical framework for understanding how local institutions can operationalize the SDGs at the sub-national level, particularly in contexts where state capacity is limited, and development outcomes rely heavily on community-based governance arrangements (Nugraheni et al., 2025).

While this study employed purposive sampling across six VOE within Indonesia, the findings should be interpreted as contextually embedded rather than universally generalisable. Indonesia provides a distinctive institutional environment characterised by decentralised village governance and legally formalised VOE. As context-sensitive research suggested, governance dynamics were often shaped by institutional, cultural, and regulatory configurations (Blanco et al., 2022; George et al., 2016). Therefore, this study aims to develop an analytically transferable framework rather than a statistically generalisable model.

Regarding methodological robustness, the AHP consistency ratio (CR = 0.06) met the acceptable threshold (<0.1). In addition, cross-validation was achieved through methodological triangulation. The quantitative AHP prioritisation was interpreted alongside thematic interview findings and field observations. Mixed-methods literature suggested that such integration enhanced internal validity by reducing single-method bias (Molina-Azorin, 2012; Schoonenboom & Johnson, 2017). While geographically bounded, the study maintained methodological rigor through consistency testing, sensitivity reflection, and triangulated validation. Consistency testing has implications for the development of a robust and replicable governance model, as it is based on the

relevance of research findings. Furthermore, sensitivity analysis helps identify key variables that influence implementation success or failure, thereby making the resulting model more adaptable to the village's social, economic, and environmental dynamics. Meanwhile, validation through data triangulation strengthens the legitimacy of research findings by ensuring that conclusions are not based on a single perspective but have been tested through multiple approaches. In practice, these three aspects contribute to the formulation of more accurate, context-specific, and sustainable village-owned enterprise management policies and strategies.

Practically, the results of this study pointed to the need for institutional reform of VOEs toward a more adaptive, open, and knowledge-based system. First, human resource capacity-building should focus on competency-based training that integrates business management and ecological literacy. The weighting results indicated that the managerial competence and environmental literacy sub-criteria had the highest weight (0.1667), indicating a strategic demand for village managers to understand the relationship between economic activity and environmental carrying capacity (Lazos-Ruiz et al., 2014).

Second, transparency and accountability should be strengthened through the digitization of public financial reports and community-based audits. This practice not only enhances the social legitimacy of VOEs (Bovens, 2007) but also fosters collective trust, which forms the basis for cross-sector collaboration (Pratiwi et al., 2024). Third, the development of strategic partnerships with the private sector and non-governmental organizations should focus on green technology transfer and access to market networks (Lockwood, 2010). This approach not only expands economic opportunities but also mitigates the limited capital and institutional capacity that have been the main obstacles to VOEs.

Fourth, this study emphasized that community participation should not be viewed as merely a formality of deliberation but rather as a social learning process that shaped collective ecological behavior. The cases of the Ponggok and Kenteng VOEs demonstrated that water resource conservation and waste management were effective only when the community was directly involved in program planning and implementation. Therefore, village community empowerment should move from an informative to a transformative level of participation (Chambers, 1997; Petty & Krosnick, 2014).

For policymakers, the findings of this study held important policy implications when situated within the regulatory framework of Indonesia's Village Law (Law No. 6 of 2014 on Villages). The Village Law mandated that village development should enhance economic self-reliance, strengthen participatory governance, and improve community welfare through locally driven institutions, including VOEs/BUMDes. In this context, the SDG-based governance model proposed in this study did not introduce an external normative agenda, but rather operationalized the core principles of the Village Law through a structured and measurable governance framework.

First, the dominance of human resource capacity in the AHP results underscored the need for institutionalized managerial certification and structured capacity-building programs for VOE administrators. Article 87 of the Village Law emphasized that VOEs had to be professionally managed to optimize village assets and improve welfare. Therefore, national and district-level policymakers should develop a tiered competency standard for VOE management, linking certification to access to village capital participation and performance-based incentives. Such institutional reinforcement would translate the legal mandate of professionalism into measurable governance indicators aligned with SDG 4 (quality education) and SDG 8 (decent work and economic growth).

Second, the prioritization of transparency and accountability supports the Village Law's provisions regarding participatory planning and financial openness in village governance. To strengthen compliance and enhance institutional trust, policymakers should integrate digital financial reporting systems into existing village information platforms. A standardized digital accounting and disclosure mechanism would not only improve oversight by village councils and communities but also reinforce SDG 16 (peace, justice, and strong institutions). More importantly, this reform should be implemented gradually to accommodate varying technological readiness across villages.

Third, in this study the typological differentiation to be identified among natural resource-dependent, eco-tourism-oriented, and hybrid VOEs suggested that a uniform governance model might be insufficient. The Village Law allows flexibility in institutional arrangements according to local potential. Governance guidelines should become adaptive to business typology. For resource-dependent VOEs, environmental safeguards and ecological carrying-capacity monitoring should be embedded in operational licensing. For eco-tourism enterprises, green certification and community co-ownership mechanisms should be encouraged. For hybrid enterprises, diversification strategies and risk management planning should be supported. Such differentiation ensures alignment with SDGs 12, 13, and 15 while remaining consistent with the principle of local autonomy in the Village Law.

Fourth, although environmental commitment and community participation received lower formal weights in the AHP hierarchy, qualitative findings revealed their critical role in long-term socio-ecological resilience. This suggested that policymakers should not rely solely on managerial prioritization but had to safeguard ecological and participatory dimensions through regulatory design. Institutionalizing annual village enterprise assemblies, participatory evaluation forums, and environmental compliance reviews would strengthen democratic accountability while maintaining coordination efficiency. This approach reflected the Village Law's participatory

ethos and advanced SDG 16 and SDG 17 (partnerships for the goals).

Finally, the findings in this study indicated that SDG-based governance should be framed not merely as administrative compliance but as a strategic transformation of VOs into socio-ecological development institutions. Therefore, rather than imposing rigid top-down regulation, policymakers should introduce incentive-based mechanisms, such as performance-linked village fund allocations, green certification rewards, and collaborative platforms at the district level to facilitate knowledge exchange and inter-village partnerships.

It is important to emphasize that these policy implications are contextually bounded within Indonesia's decentralized governance structure. The proposed model is analytically transferable to similar community enterprise contexts and should be adapted to institutional and cultural conditions in other developing countries.

Overall, aligning SDG-based governance standards with the Village Law framework enables a shift in paradigm: From viewing VOs primarily as local revenue generators to recognizing them as integrated institutions that balance economic viability, social legitimacy, and ecological responsibility in pursuit of sustainable village development.

6. Conclusions

This study revealed that sustainability-based VOs were no longer solely measured by economic aspects; instead, they should be supported by human-oriented and environmentally friendly governance. The results of the analysis using the AHP indicated that human resource capacity and quality were the most influential dimensions (weighted at 0.3333), followed by transparency and accountability (0.2667), and cross-sector collaboration and partnerships (0.2000). However, environmental commitment (0.1333) and community participation (0.0667), though numerically lower, were the “spirit” that determined the socio-ecological sustainability of the institution.

Field findings indicated that VOs that successfully survived and thrived effectively integrated business logic with ecological ethics. Ponggok VOs, for example, incorporated upstream water conservation into its water tourism business strategy, while Kenteng VOs integrated waste management with community empowerment. These practices demonstrated that village economic sustainability could only grow from collective ecological awareness and not simply managerial efficiency.

Conceptually, this research marked a paradigm shift in VOs from economic entities to socio-ecological institutions that continuously learnt and adapted continuously. Human resources are no longer positioned as administrative implementers but rather as learning agents that build values, collaboration, and social legitimacy. Practically, this environment-based governance model offers a new direction for village policy reform toward an inclusive, transparent, and resilient green economic system.

Thus, this study confirmed that environmental-based VO governance was not merely a technical strategy but a paradigmatic transformation from a short-term profit orientation to a balance between welfare, social justice, and ecological sustainability. When villages are able to manage their resources intelligently, transparently, and from a green perspective, VOs are not merely the engine of the village economy, but also the guardian of harmony between humans and nature, as well as the driving force behind achieving the SDGs from the roots of national development.

7. Recommendations for Further Research

This research developed a sustainability-based VO governance model utilizing the AHP approach, with a focus on human resource capacity, transparency, collaboration, and ecological commitment. However, the complex social, political, and cultural dynamics of villages require a comprehensive and participatory follow-up approach to deepen the validity of the results and to broaden the context of their application.

First, future research should use the Delphi method to test and validate the resulting governance model. This approach enables researchers to gather opinions from cross-disciplinary experts, including academics, village practitioners, and policymakers, to reach consensus based on reflection and iteration (Turoff & Linstone, 1975). By involving experts' judgment, the Delphi process could strengthen the robustness of the governance model developed by AHP, particularly in determining policy priorities and sustainability indicators contextual to the village. The combined AHP–Delphi approach has been demonstrated to enhance decision accuracy in studies on local resource governance and sustainable development (Rombe et al., 2021; Saadati et al., 2025).

Second, future research should expand the unit of analysis to include more VOs with varying geographical, social, and economic characteristics. This study utilized six sustainability-based VOs; however, the context across regions in Indonesia was highly heterogeneous. Expanding the sample size would provide a more representative understanding of variations in governance practices and levels of ecological literacy across villages. This approach aligns with the recommendations of Lazos-Ruiz et al. (2014), who emphasized the importance of considering differences in socio-ecological contexts in developing adaptive green rural enterprises.

Third, future research should integrate a mixed-methods approach by combining quantitative analysis, such as AHP or Structural Equation Modeling (SEM), with qualitative data based on institutional ethnography. This

approach captures social and cultural dynamics that cannot be measured numerically, such as social trust, local leadership, and village social capital (Kakumba, 2010). This integration enhances the theoretical understanding of the relationship between institutional structures and grassroots sustainability practices.

Fourth, the digital governance aspects require further research. Digitizing the financial reporting and operational management systems of VOs has the potential to strengthen transparency and accountability, as emphasized by Bovens (2007), while simultaneously expanding citizen engagement through open data access. Empirical studies on the application of digital technology in village governance are still limited; therefore, research in this area will make a significant contribution to the modernization of VOs in the era of digital transformation.

Finally, further research is required to deepen the integration of environmental and social dimensions within the eco-governance framework. While environmental commitment ranked fourth in this study based on AHP priority weighting, future studies should investigate the causal relationship between ecological literacy, collaboration, and village economic sustainability using a mediation or moderation analysis model. This approach aligns with the study by Chaaben et al. (2024), who asserted that the quality of the interaction between environmental policies, human capacity, and institutional governance determined long-term economic sustainability.

Future research should focus on strengthening participatory and interdisciplinary methodologies to develop more contextual, adaptive, and scalable VO governance models. A Delphi process-based approach, combined with the development of a mixed-methods governance model, will enrich scientific understanding and strengthen the role of VOs as a catalyst for sustainable development at the village level.

Author Contributions

Conceptualization, B.D.H., N.S., and M.E.S.; methodology, B.D.H.; software, N.S.; validation, B.D.H., M.E.S., and N.A.; formal analysis, S.M.; investigation, N.A.; resources, B.D.H.; data curation, N.A.; writing—original draft preparation, B.D.H.; writing—review and editing, M.E.S.; visualization, N.S.; supervision, B.D.H.; project administration, S.M. and N.A.; funding acquisition, B.D.H. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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